Westchester County Sidewalk Coverage Assessment

Metro-North Transit Station Area Analysis
Technical Analysis Report

Arcanum Research Initiative

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1 Executive Summary

Analysis Objective: Quantify sidewalk coverage adequacy on roads within 0.5 miles of Metro-North train stations in Westchester County using the DVRPC (Delaware Valley Regional Planning Commission) sidewalk-to-road ratio methodology.

Key Findings:

- **TOD Area Coverage:** 54.9% of roads within 0.5 miles of Metro-North stations have sidewalk coverage (615 out of 1,117 roads)
- **TOD vs Non-TOD Gap:** TOD areas have 3.0x better coverage than non-TOD areas (54.9% vs 18.2%)
- **County-Wide Context:** Only 26.0% of all county roads have sidewalk coverage (1,141 out of 4,386 roads)
- **Coverage Inequality:** 74.0% of county roads (3,245 roads) have no sidewalk coverage on either side

Policy Implications: The analysis reveals concentrated pedestrian infrastructure investment near transit stations, with TOD areas receiving substantially better sidewalk coverage. However, 45.1% of TOD roads still lack sidewalks, indicating opportunity for targeted improvement.

Statistical Confidence: Analysis covers 100% of county road polygons (4,386 roads, 15,514.1 acres) with spatially validated geometries using EPSG:2260 (NY State Plane Long Island, feet) projected coordinate system.

2 Data Overview

2.1 Data Sources and Coverage

Data Source	Features	Spatial Extent
County Roadways Polygon	4,386 roads	15,514.1 acres
County Sidewalks Polygon	Multiple polygons	Spatially matched
Metro-North Stations	Multiple stations	Point locations

Table 1: Data Sources and Coverage

Coordinate System: EPSG:2260 (NY State Plane Long Island, feet) - Projected CRS optimized for accurate distance calculations in Westchester County

Buffer Distance: 0.5 miles (2,640 feet) - Industry standard for Transit-Oriented Development (TOD) analysis

Analysis Period: October 2025

2.2 Data Quality Assessment

- **Completeness:** 100% of county road polygons analyzed
- **Geometric Validation:** All geometries validated as valid polygons with no topology errors

- **Spatial Accuracy:** Projected CRS ensures accurate distance calculations (feet units)
- **Attribute Completeness:** Road type classifications available for all roads
- **Temporal Currency:** County-provided shapefiles represent current infrastructure

2.3 Data Processing Pipeline

1. **Data Import:** Load road polygons, sidewalk polygons, and Metro-North station points
2. **Coordinate System Validation:** Ensure EPSG:2260 projection for all layers
3. **TOD Buffer Generation:** Create 0.5-mile (2,640 feet) buffers around Metro-North stations
4. **Spatial Indexing:** Build R-tree spatial index for efficient polygon intersection queries
5. **Coverage Analysis:** Apply DVRPC methodology to each road polygon
6. **Statistical Aggregation:** Compute county-wide and TOD-specific statistics
7. **Output Generation:** Export GeoJSON files (EPSG:4326), Excel reports, and JSON statistics

3 Methodology

3.1 DVRPC Sidewalk-to-Road Ratio Approach

The analysis implements the DVRPC (Delaware Valley Regional Planning Commission) sidewalk-to-road ratio methodology, a city planning standard for assessing pedestrian infrastructure coverage. This approach provides superior accuracy compared to simple centroid-based or intersection methods.

Core Principle: For each road polygon, calculate the ratio of sidewalk length to road perimeter on each side (left and right) to determine coverage type.

3.1.1 Algorithm Steps

For each road polygon:

1. **Centerline Extraction:** Compute road centerline using polygon skeleton algorithm
2. **Edge Buffer Generation:** Create 5-foot offset buffers from left and right road edges
3. **Sidewalk Detection:** Identify sidewalk polygons that intersect with edge buffers
4. **Ratio Calculation:** Compute sidewalk-to-road perimeter ratios for left and right sides
5. **Coverage Classification:** Apply threshold-based classification:
    - **No Coverage:** Both sides have ratio < 0.1
    - **One-Side Coverage:** One side has ratio in range $[0.4, 0.8]$, other side < 0.1
    - **Both-Sides Coverage:** Both sides have ratio > 1.2

3.2 Coverage Classification Thresholds

Coverage Type	Left Side Ratio	Right Side Ratio
No Coverage	< 0.1	< 0.1
One-Side (Left)	≥ 0.4	< 0.1
One-Side (Right)	< 0.1	≥ 0.4
Both-Sides	> 1.2	> 1.2

Table 2: DVRPC Coverage Classification Thresholds

Threshold Rationale:

- **0.1 (No Coverage):** Filters out incidental sidewalk fragments and measurement noise
- **0.4–0.8 (One-Side):** Captures meaningful partial coverage where one side has substantial sidewalk
- **1.2 (Both-Sides):** Ensures both sides have continuous sidewalk infrastructure (ratio > 1.0 accounts for sidewalks extending beyond strict road edges)

3.3 TOD Buffer Analysis

Buffer Standard: 0.5 miles (2,640 feet) - Half-mile pedestrian catchment area is the industry standard for TOD analysis, representing comfortable walking distance to transit stations.

Spatial Operations:

1. Create 0.5-mile buffers around Metro-North station point locations
2. Merge overlapping buffers to create unified TOD zones
3. Identify all road polygons whose centroids fall within TOD buffers
4. Classify roads as "TOD" or "Non-TOD" based on buffer intersection

4 County-Wide Results

4.1 Overall Coverage Statistics

Coverage Type	Road Count	Percentage
No Coverage (neither side)	3,245	74.0%
One-Side Coverage	620	14.1%
Both-Sides Coverage	521	11.9%
ANY Coverage (Total)	**1,141**	**26.0%**

Table 3: County-Wide Sidewalk Coverage Breakdown

Key Observation: Nearly three-quarters of county roads (74.0%) have no sidewalk coverage on either side, indicating substantial gaps in pedestrian infrastructure.

4.2 Coverage by Road Area

Coverage Type	**Road Area (Acres)**	**Percentage**
No Coverage	4,239.5	27.3%
One-Side Coverage	10,753.9	69.3%
Both-Sides Coverage	520.8	3.4%

Table 4: County-Wide Coverage by Road Area

Area vs Count Discrepancy: While 74.0% of roads by count lack coverage, only 27.3% of road area lacks coverage. This indicates that larger roads (by area) have better sidewalk coverage than smaller roads.

4.3 Coverage by Road Type

Road Type	**Total Roads**	**Coverage %**	**No Coverage**
Major Paved Alley	28	57.1%	12
Hidden Road	816	51.5%	396
Elevated Road	1,262	30.4%	878
Paved Road	937	28.4%	671
Elevated Hidden Road	13	15.4%	11
Unpaved Road	1,330	4.0%	1,277

Table 5: Sidewalk Coverage by Road Type (Sorted by Coverage %)

Key Insights:

- **Major Paved Alleys** show highest coverage (57.1%) but represent small sample (28 roads)
- **Hidden Roads** have surprisingly strong coverage (51.5%) - 420 out of 816 roads
- **Unpaved Roads** show extremely low coverage (4.0%) - only 53 out of 1,330 roads
- **Paved Roads** and **Elevated Roads** have similar moderate coverage (28.4% and 30.4%)

5 Transit-Oriented Development (TOD) Analysis

5.1 TOD Area Coverage Assessment

Coverage Type	**Road Count**	**Percentage**
No Coverage (neither side)	502	45.1%
One-Side Coverage	352	31.5%
Both-Sides Coverage	263	23.5%
ANY Coverage (Total)	**615**	**54.9%**

Table 6: TOD Area Sidewalk Coverage (0.5-Mile Buffers)

Answer to Taylor's Question: 54.9% of roads within 0.5 miles of Metro-North stations have sidewalk coverage (615 out of 1,117 total roads). This represents MODERATE ADEQUACY with substantial room for improvement.

5.2 TOD vs Non-TOD Comparison

Area Type	Coverage %	Total Roads	With Coverage	No Coverage
TOD Areas (0.5 mi)	54.9%	1,117	615	502
Non-TOD Areas	18.2%	3,269	526	2,743
County-Wide	26.0%	4,386	1,141	3,245

Table 7: TOD vs Non-TOD Coverage Comparison

Coverage Ratio: TOD areas have 3.0x better coverage than non-TOD areas (54.9% vs 18.2%)

Statistical Significance: The 36.7 percentage point gap between TOD and non-TOD areas indicates substantial, targeted investment in pedestrian infrastructure near transit stations.

5.3 TOD Coverage Breakdown by Type

Coverage Type	TOD Areas		Non-TOD Areas	
	Count	%	Count	%
No Coverage	502	45.1%	2,743	83.9%
One-Side Coverage	352	31.5%	268	8.2%
Both-Sides Coverage	263	23.5%	258	7.9%

Table 8: Coverage Type Breakdown: TOD vs Non-TOD

Key Observations:

- TOD areas have far more one-side coverage (31.5% vs 8.2%) - likely partial build-out in progress
- TOD areas have 3x more both-sides coverage (23.5% vs 7.9%) - complete pedestrian corridors
- Non-TOD areas are dominated by no coverage (83.9%) - severely underserved

6 Detailed Findings

6.1 TOD Area Pedestrian Infrastructure Gap

Finding: Despite 3.0x better coverage than non-TOD areas, 45.1% of TOD roads (502 roads) still lack sidewalks on either side.

Evidence:

- 502 out of 1,117 TOD roads have no sidewalk coverage
- This represents a pedestrian connectivity gap within comfortable walking distance of transit

- 352 additional roads have only one-side coverage (opportunity for completion)

Implications:

- **Transit Access Barriers:** Residents may face unsafe or inaccessible walking routes to Metro-North stations
- **Investment Opportunity:** Completing 502 no-coverage roads would raise TOD coverage to 100%
- **One-Side Completion:** Completing 352 one-side roads would provide full both-sides access

6.2 Non-TOD Area Coverage Inequality

Finding: Non-TOD areas have only 18.2% coverage, with 83.9% of roads (2,743 roads) having no sidewalks on either side.

Evidence:

- 2,743 out of 3,269 non-TOD roads lack sidewalk coverage
- Only 526 non-TOD roads have any sidewalk coverage
- Non-TOD coverage is 3.0x worse than TOD areas

Implications:

- **Pedestrian Access Equity:** Residents outside TOD areas face substantially worse walkability
- **Car Dependency:** Lack of sidewalks forces automobile dependency in non-TOD areas
- **Public Health Impact:** Limited pedestrian infrastructure reduces opportunities for active transportation

6.3 Road Type Coverage Disparities

Finding: Unpaved roads show extremely low coverage (4.0%) while hidden roads show surprisingly high coverage (51.5%).

Evidence:

- Unpaved roads: 1,277 out of 1,330 lack coverage (96.0% no coverage rate)
- Hidden roads: 420 out of 816 have coverage (51.5% coverage rate)
- Major paved alleys: 16 out of 28 have coverage (57.1% coverage rate)

Implications:

- **Unpaved Road Neglect:** Unpaved roads are systematically excluded from sidewalk investment
- **Hidden Road Paradox:** "Hidden roads" (likely residential or service roads) receive relatively good coverage
- **Road Classification Matters:** Sidewalk investment decisions appear strongly influenced by road type

7 Quality Assurance and Validation

7.1 Spatial Validation

Geometric Validation: All 4,386 road polygons validated as topologically correct with no invalid geometries

Coordinate System Accuracy: EPSG:2260 (NY State Plane Long Island, feet) ensures accurate distance calculations for:

- Road perimeter measurements
- Sidewalk length measurements
- TOD buffer generation (0.5 miles = 2,640 feet)
- Edge offset buffers (5 feet from road edges)

Spatial Index Performance: R-tree spatial indexing enables efficient polygon intersection queries (O(log n) complexity)

7.2 Methodology Validation

DVRPC Methodology: Industry-standard approach used by city planning departments for pedestrian infrastructure assessment

Threshold Sensitivity: Coverage thresholds (0.1, 0.4–0.8, 1.2) validated against visual inspection of sample roads

Edge Detection Accuracy: 5-foot offset buffers capture sidewalks along road edges while excluding distant sidewalks

7.3 Output Validation

Data Integrity Checks:

- Sum of coverage categories equals total roads (3,245 + 620 + 521 = 4,386)
- TOD + Non-TOD roads sum to county total (1,117 + 3,269 = 4,386)
- All percentage calculations verified (e.g., 1,141 / 4,386 = 26.0%)
- GeoJSON exports validated in QGIS (EPSG:4326 for web compatibility)

8 Policy Recommendations

8.1 Immediate Priority Actions

1. **TOD No-Coverage Gap Closure:** Prioritize sidewalk installation on 502 TOD roads with no coverage to maximize transit connectivity
2. **TOD One-Side Completion:** Complete 352 TOD roads with one-side coverage to achieve full both-sides pedestrian access
3. **Unpaved Road Targeting:** Develop sidewalk installation strategy for unpaved roads (currently 4.0% coverage)

8.2 Strategic Planning Recommendations

- **Equity Analysis:** Investigate socioeconomic factors associated with TOD vs non-TOD coverage gap
- **Transit Access Study:** Map pedestrian routes from residential areas to Metro-North stations
- **ADA Compliance Assessment:** Evaluate sidewalk quality and accessibility beyond binary coverage
- **Multi-Modal Integration:** Coordinate sidewalk expansion with bike lane and bus route planning

8.3 Long-Term Vision

75% TOD Coverage Target: Achieving 75% coverage in TOD areas would require completing 224 additional roads (from current 615 to 839 roads with coverage)

50% County-Wide Target: Raising county-wide coverage to 50% would require adding sidewalks to 1,052 roads (from 1,141 to 2,193 roads with coverage)

Non-TOD Area Focus: Closing the TOD vs non-TOD gap would require adding sidewalks to 1,178 non-TOD roads (to reach 54.9% parity)

9 Limitations and Future Enhancements

9.1 Data Limitations

- **Binary Coverage:** Analysis classifies coverage as present/absent but does not assess sidewalk quality, width, or ADA compliance
- **Temporal Snapshot:** County shapefiles represent a single point in time; does not track infrastructure change over time
- **Network Connectivity:** Does not analyze whether sidewalks form continuous pedestrian networks to key destinations
- **Socioeconomic Context:** Lacks demographic or economic data to assess equity implications

9.2 Methodological Considerations

- **Threshold Sensitivity:** Coverage classification depends on ratio thresholds (0.1, 0.4–0.8, 1.2) - alternative thresholds may yield different results
- **Edge Buffer Size:** 5-foot offset buffers may miss sidewalks set back from road edges or capture unrelated infrastructure
- **TOD Buffer Distance:** 0.5-mile buffer is standard but actual walking catchment varies by topography, road crossings, and other barriers

9.3 Future Enhancement Opportunities

1. **Network Analysis:** Assess pedestrian connectivity from residential areas to transit stations, schools, commercial centers
2. **Quality Assessment:** Integrate sidewalk condition data (surface quality, width, ADA compliance, lighting)
3. **Temporal Analysis:** Track sidewalk coverage changes over time using historical shapefiles
4. **Equity Analysis:** Overlay demographic data to identify underserved populations
5. **Multi-Modal Integration:** Combine sidewalk coverage with bike lane, bus route, and crosswalk data
6. **Predictive Modeling:** Develop models to prioritize sidewalk investments based on demand, cost, and equity factors

10 Data and Code Availability

10.1 Machine-Readable Datasets (Excel)

Location: `D:/Arcanum/Projects/Westchester/Output/Excel/`

- `1_EXECUTIVE_SUMMARY.xlsx` - Answer front and center with complete breakdown (ONE SHEET)
- `2_TOD_COMPARISON.xlsx` - Side-by-side TOD vs Non-TOD analysis (ONE SHEET)
- `3_ROAD_TYPE_ANALYSIS.xlsx` - Coverage by road type classification (ONE SHEET)
- `4_AREA_ANALYSIS.xlsx` - Coverage statistics by area in acres (ONE SHEET)

Note: All Excel files follow Druck standard with ONE SHEET per file.

10.2 Geospatial Outputs (GeoJSON)

Location: `D:/Arcanum/Projects/Westchester/Technical/data/processed/countywide_sidewalk_analysis`

County-Wide Files (EPSG:4326):

- `roads_no_coverage.geojson` - 3,245 roads without sidewalks
- `roads_one_side.geojson` - 620 roads with one-side coverage
- `roads_both_sides.geojson` - 521 roads with both-sides coverage

TOD-Specific Files (EPSG:4326):

- `tod_roads_no_coverage.geojson` - 502 TOD roads without sidewalks
- `tod_roads_one_side.geojson` - 352 TOD roads with one-side coverage
- `tod_roads_both_sides.geojson` - 263 TOD roads with both-sides coverage
- `metro_north_buffers.geojson` - 0.5-mile TOD buffer zones

Total GeoJSON Size: 596 MB (web-ready EPSG:4326 format for QGIS, ArcGIS, Leaflet, Mapbox)

10.3 Statistical Summaries (JSON)

Location: `D:/Arcanum/Projects/Westchester/Technical/data/processed/countywide_sidewalk_analysis/`

- `county_wide_statistics.json` - Complete county-wide metrics including coverage counts, percentages, area statistics, and road type breakdowns
- `tod_statistics.json` - TOD vs Non-TOD comparison metrics with detailed breakdowns

10.4 Analysis Code

Location: `D:/Arcanum/Projects/Westchester/Technical/src/`

Core Analysis Scripts:

- `countywide_sidewalk_analyzer.py` - Main analysis engine (652 lines) implementing DVRPC methodology
- `transit_sidewalk_analyzer.py` - TOD buffer analysis and TOD vs non-TOD comparison
- `generate_excel_reports.py` - Druck-compliant Excel report generator

Reproduction Instructions: See `Technical/README.md` for step-by-step execution guide

11 Conclusion

Primary Answer: 54.9% of roads within 0.5 miles of Metro-North stations have sidewalk coverage (615 out of 1,117 roads), representing MODERATE ADEQUACY with substantial room for improvement.

Key Insights:

- TOD areas have 3.0x better coverage than non-TOD areas (54.9% vs 18.2%)
- 45.1% of TOD roads (502 roads) still lack sidewalks - priority gap for transit access
- 83.9% of non-TOD roads lack coverage - severe pedestrian infrastructure inequality
- Unpaved roads show only 4.0% coverage - systematically underserved road type

Research Contribution: This analysis provides the first comprehensive spatial assessment of sidewalk coverage in Westchester County using industry-standard DVRPC methodology with 1-side vs 2-side detection.

Practical Value: Results enable data-driven prioritization of sidewalk investments to maximize transit connectivity, pedestrian safety, and infrastructure equity.